

73. Auto-Negotiation for backplane and copper cable assembly

Auto-Negotiation, as defined in this clause, is specified for use with Ethernet PHYs operating over a backplane and for use with certain Ethernet PHYs operating over a copper cable assembly.

73.1 Auto-Negotiation introduction

While implementation of Auto-Negotiation is mandatory for Backplane Ethernet PHYs, the use of Auto-Negotiation is optional. Parallel detection shall be provided for legacy devices that do not support Auto-Negotiation.

The Auto-Negotiation function allows an Ethernet device to advertise modes of operation it possesses to another device at the remote end of a Backplane Ethernet link and to detect corresponding operational modes the other device may be advertising.

The objective of this Auto-Negotiation function is to provide the means to exchange information between two devices that share a link across a backplane and to automatically configure both devices to take maximum advantage of their abilities. It has the additional objective of supporting a digital signal detect to ensure that the device is attached to a link partner rather than detecting signal due to crosstalk.

Auto-Negotiation is performed using differential Manchester encoding (DME) pages. DME provides a DC-balanced signal. DME does not add packet or upper layer overhead to the network devices.

Auto-Negotiation does not test the link segment characteristics.

This function allows the devices at both ends of a link segment to advertise abilities, acknowledge receipt and discover the common modes of operation that both devices share, and to reject the use of operational modes that are not shared by both devices. Where more than one common mode exists between the two devices, a mechanism is provided to allow the devices to resolve to a single mode of operation using a predetermined priority resolution function. The Auto-Negotiation function allows the devices to switch between the various operational modes in an orderly fashion, permits management to disable or enable the Auto-Negotiation function, and allows management to select a specific operational mode. The Auto-Negotiation function also provides a parallel detection function to allow Backplane Ethernet devices to connect to other Backplane Ethernet devices that have Auto-Negotiation disabled and interoperate with legacy devices that do not support Clause 73 Auto-Negotiation.

It is recommended that a device that has negotiated 1000BASE-KX operation through this clause not perform Clause 37 Auto-Negotiation. A device that performs Clause 37 Auto-Negotiation after having negotiated 1000BASE-KX operation through Clause 73 Auto-Negotiation will not interoperate with a device that does not perform Clause 37 Auto-Negotiation. Therefore, a device that intends to enable Clause 37 Auto-Negotiation after Clause 73 Auto-Negotiation has completed shall ensure through an implementation-specific mechanism that the link partner supports Clause 37 Auto-Negotiation and intends to enable it. If Clause 37 Auto-Negotiation is performed after Clause 73 Auto-Negotiation, then the advertised abilities used in the Clause 37 Auto-Negotiation shall match those advertised abilities used in the Clause 73 Auto-Negotiation.

The Auto-Negotiation functions are listed in 73.3.

73.2 Relationship to the ISO/IEC Open Systems Interconnection (OSI) reference model

The Auto-Negotiation function is provided at the Physical Layer of the ISO/IEC OSI reference model as shown in Figure 73–1. A device that supports multiple modes of operation may advertise its capabilities using the Auto-Negotiation function. The actual transfer of information is observed only at the MDI or on the backplane medium.

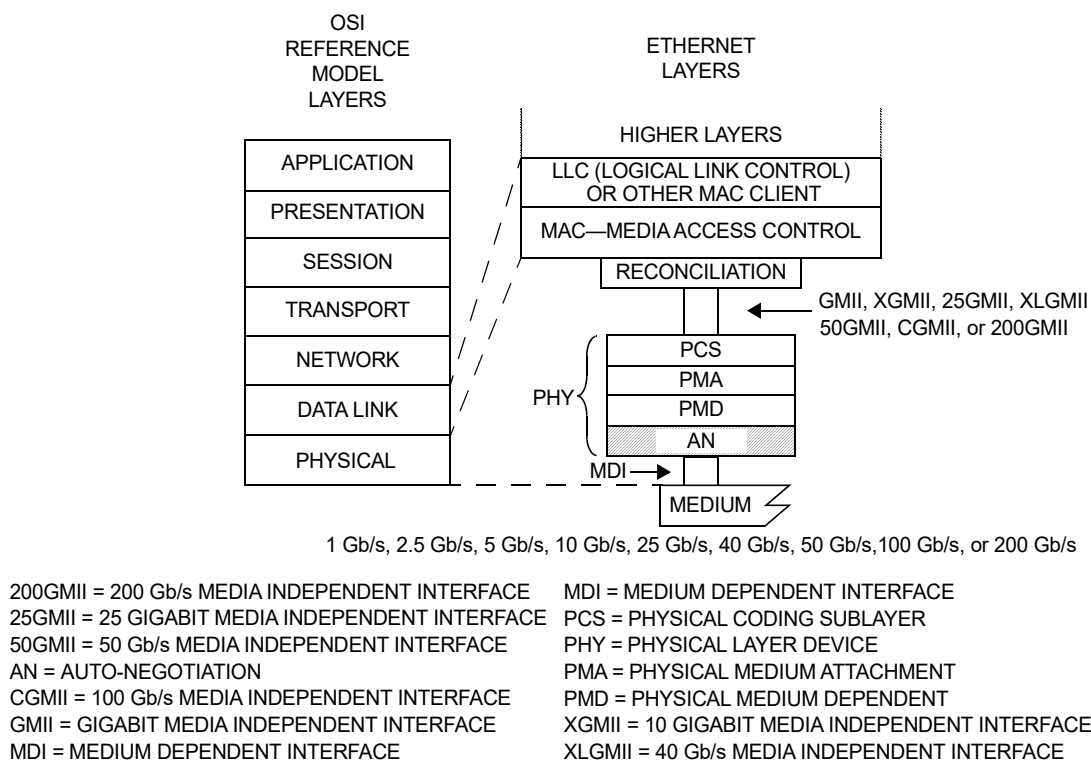


Figure 73–1—Location of Auto-Negotiation function within the ISO/IEC OSI reference model

73.3 Functional specifications

The Auto-Negotiation function provides a mechanism to control connection of a single MDI to a single PHY type, where more than one PHY type may exist. A management interface provides control and status of Auto-Negotiation, but the presence of a management agent is not required.

The Auto-Negotiation function shall provide the following:

- Auto-Negotiation transmit
- Auto-Negotiation receive
- Auto-Negotiation arbitration

These functions shall comply with the state diagrams from Figure 73–9 through Figure 73–11. The Auto-Negotiation functions shall interact with the technology-dependent PHYs through the

Technology-Dependent interface (see 73.9). Technology-Dependent PHYs are those supported by the Auto-Negotiation process (see Table 73–4).

When the MDI supports multiple lanes, then lane 0 of the MDI shall be used for Auto-Negotiation and for connection of any single-lane PHYs (e.g., 1000BASE-KX or 10GBASE-KR).

73.4 Transmit function requirements

The Transmit function provides the ability to transmit pages. The first pages exchanged by the local device and its link partner after Power-On, link restart, or renegotiation contain the base link codeword defined in Figure 73–6. The local device may modify the link codeword to disable an ability it possesses, but will not transmit an ability it does not possess. This makes possible the distinction between local abilities and advertised abilities so that multi-ability devices may Auto-Negotiate to a mode lower in priority than the highest common ability.

73.5 DME transmission

Auto-Negotiation’s method of communication builds upon the encoding mechanism known as differential Manchester encoding (DME). The DME page encodes the data that is used to control the Auto-Negotiation function. DME pages shall not be transmitted when Auto-Negotiation is complete and the highest common denominator PHY has been enabled.

73.5.1 DME electrical specifications

Transmitter characteristics shall meet the specifications in Table 73–1 at TP1 while transmitting DME pages. Receiver characteristics shall meet the specifications in Table 73–1 at TP4 while receiving DME pages.

Table 73–1—DME electrical characteristics

Parameter	Value	Units
Transmit differential peak-to-peak output voltage	600 to 1200	mV
Receive differential peak-to-peak input voltage	200 to 1200	mV

For any multi-lane PHY, DME pages shall be transmitted only on lane 0. The transmitters on other lanes should be disabled as specified in 71.6.7, 84.7.7, 85.7.7, 92.7.7, 93.7.7, 94.3.6.7, 136.8.7, or 137.8.7.

73.5.2 DME page encoding

A DME page carries a 48-bit Auto-Negotiation page. It consists of 106 evenly spaced transition positions that contain a Manchester violation delimiter, the 48-bit page, and a single pseudo-random bit. The odd-numbered transition positions represent clock information. The even numbered transition positions represent data information. DME pages are transmitted continuously without any idle or gap.

The first eight transition positions contain the Manchester violation delimiter, which marks the beginning of the page. The Manchester violation contains a transition at position 1 and position 5 and no transitions at the remaining positions. The Manchester violation delimiter is the only place where four intervals occur between transitions. This allows the receiver to obtain page synchronization.

Each of the remaining 49 odd-numbered transition positions shall contain a transition. The remaining 49 even-numbered transition positions shall represent data information as follows:

- A transition present in an even-numbered transition position represents a logical one
- A transition absent from an even-numbered transition position represents a logical zero

The first 48 of these positions shall carry the data of the Auto-Negotiation page. The final position carries the pseudo-random bit. The value of the pseudo-random bit shall be derived from a pseudo-random generator as shown in Figure 73–2.

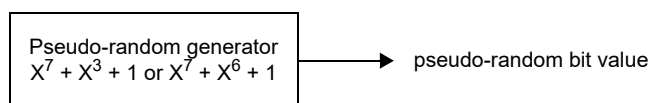


Figure 73–2—DME page bit 49 randomizer

The counter shall increment once per DME page.

The purpose of the 49th bit is to remove the spectral peaks that would otherwise occur when sending the same AN page repeatedly. Randomly choosing between 0 or 1 for one of the DME bits results in randomly inverting or not inverting the encoded page so that repetitions of the same page no longer produce a periodic signal.

Clock transition positions are differentiated from data transition positions by the spacing between them, as shown in Figure 73–3 and enumerated in Table 73–2.

The encoding of data using DME bits in an DME page is illustrated in Figure 73–3.

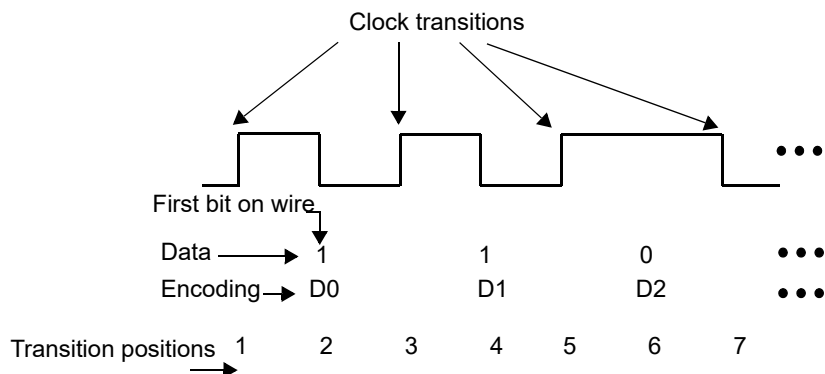


Figure 73–3—Data bit encoding within DME pages

73.5.3 DME page timing

The timing parameters for DME pages shall be followed as in Table 73–2. The transition positions within a DME page are spaced with a period of T1. T2 is the separation between clock transitions. T3 is the time from a clock transition to a data transition representing a one. The period, T1, shall be 3.2 ns ± 0.01%. Transitions shall occur within ± 0.2 ns of their ideal positions.

T5 specifies the duration of a DME page. Since DME pages are sent continuously during Auto-Negotiation, T5 is also the time from the start of one DME page to the start of the next DME page.

The minimum number of transitions and maximum number of transitions in a page is represented by T4.

Table 73–2 summarizes the timing parameters. The transition timing parameters are illustrated in Figure 73–4.

Table 73–2— DME page timing summary

	Parameter	Min.	Typ.	Max.	Units
T1	Transition position spacing (period)	3.2 –0.01%	3.2	3.2 +0.01%	ns
T2	Clock transition to clock transition	6.2	6.4	6.6	ns
T3	Clock transition to data transition (data = 1)	3.0	3.2	3.4	ns
T4	Transitions in a DME page	51	—	100	—
T5	DME page width	338.8	339.2	339.6	ns
T6	DME Manchester violation delimiter width	12.6	12.8	13.0	ns

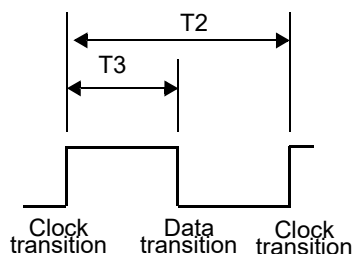


Figure 73–4—DME page transition timing

73.5.3.1 Manchester violation delimiter

A violation is signaled as shown in Figure 73–5.

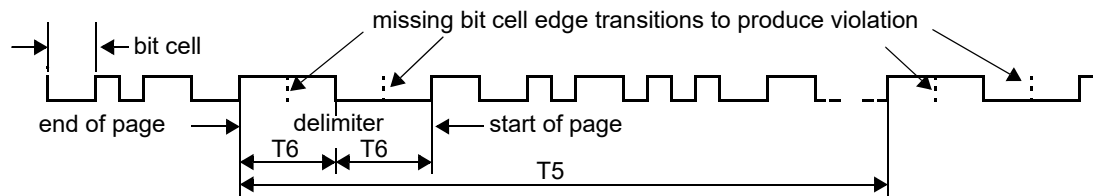


Figure 73–5—Manchester violation

73.6 Link codeword encoding

The base link codeword (Base Page) transmitted within a DME page shall convey the encoding shown in Figure 73–6. The Auto-Negotiation function supports additional pages using the Next Page function. Encoding for the link codeword(s) used in the Next Page exchange are defined in 73.7.7. In a DME page, D0 shall be the first bit transmitted.

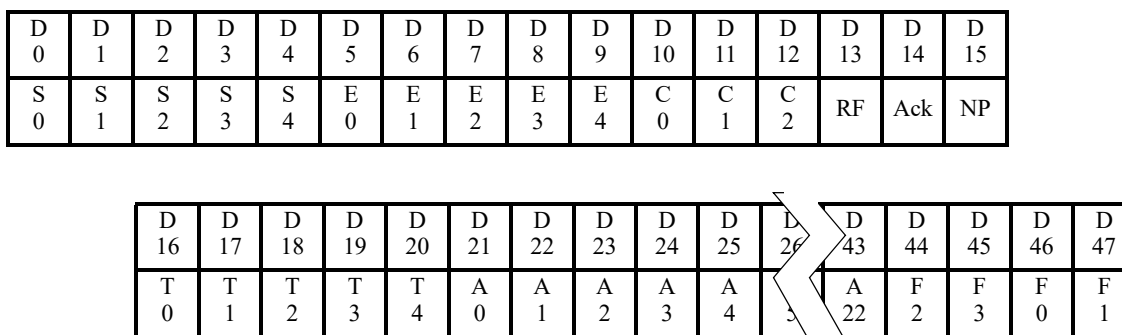


Figure 73–6—Link codeword Base Page

D[4:0] contains the Selector Field. D[9:5] contains the Echoed Nonce field. D[12:10] contains capability bits to advertise capabilities not related to the PHY. C[1:0] is used to advertise pause capability. The remaining capability bit C[2] is reserved. D[15:13] contains the RF, Ack, and NP bits. These bits shall function as specified in 28.2.1.2. D[20:16] contains the Transmitted Nonce field. D[43:21] contains the Technology Ability Field. D[47:44] contains FEC capability (see 73.6.5).

73.6.1 Selector Field

Selector Field (S[4:0]) is a five-bit wide field, encoding 32 possible messages. Selector Field encoding definitions are shown in Annex 28A. Combinations not specified are reserved for future use. Reserved combinations of the Selector Field shall not be transmitted.

The Selector Field for IEEE Std 802.3 is shown in Table 73–3.

Table 73–3—Selector Field Encoding

S4	S3	S2	S1	S0	Selector description
0	0	0	0	1	IEEE Std 802.3

73.6.2 Echoed Nonce Field

Echoed Nonce Field (E[4:0]) is a 5-bit wide field containing the nonce received from the link partner. When Acknowledge is set to logical zero, the bits in this field shall contain logical zeros. When Acknowledge is set to logical one, the bits in this field shall contain the value received in the Transmitted Nonce Field from the link partner.

73.6.3 Transmitted Nonce Field

Transmitted Nonce Field (T[4:0]) is a 5-bit wide field containing a random or pseudo-random number. A new value shall be generated for each entry to the Ability Detect state. The method of generating the nonce is left to the implementer. The transmitted nonce should have a uniform distribution in the range from 0 to $2^5 - 1$. The method used to generate the value should be designed to minimize correlation to the values generated by other devices.

73.6.4 Technology Ability Field

Technology Ability Field (A[22:0]) is a 23-bit wide field containing information indicating supported technologies specific to the selector field value when used with the Auto-Negotiation for backplane and copper cable assembly. These bits are mapped to individual technologies such that abilities are advertised in parallel for a single selector field value. The Technology Ability Field encoding for the IEEE 802.3 selector with Auto-Negotiation for backplane and copper cable assembly is described in Table 73–4.

Table 73–4—Technology Ability Field encoding

Bit	Technology
A0	1000BASE-KX
A1	10GBASE-KX4
A2	10GBASE-KR
A3	40GBASE-KR4
A4	40GBASE-CR4
A5	100GBASE-CR10
A6	100GBASE-KP4
A7	100GBASE-KR4
A8	100GBASE-CR4
A9	25GBASE-KR-S or 25GBASE-CR-S
A10	25GBASE-KR or 25GBASE-CR
A11	2.5GBASE-KX
A12	5GBASE-KR
A13	50GBASE-KR or 50GBASE-CR
A14	100GBASE-KR2 or 100GBASE-CR2
A15	200GBASE-KR4 or 200GBASE-CR4
A16 through A22	Reserved

Multiple technologies may be advertised in the link codeword. A device shall support the data service ability for a technology it advertises. It is the responsibility of the Arbitration function to determine the common mode of operation shared by a link partner and to resolve multiple common modes.

NOTE—Previous editions of this standard prohibited simultaneous advertisement of PHYs that support operation over electrical backplanes with PHYs that support operation over copper cable assemblies.

25GBASE-KR-S abilities are a subset of 25GBASE-KR abilities, and likewise 25GBASE-CR-S abilities are a subset of 25GBASE-CR abilities. To allow interoperation between 25GBASE-KR-S and 25GBASE-KR PHY types, and between 25GBASE-CR-S and 25GBASE-CR PHY types, a device that supports 25GBASE-KR or 25GBASE-CR should advertise both A9 and A10 ability bits during auto-negotiation.

The fields A[22:16] are reserved for future use. Reserved fields shall be sent as zero and ignored on receive.

73.6.5 FEC capability

FEC (F2, F3, F0, F1) is encoded in bits D44:D47 of the base link codeword. The four FEC bits are used as follows:

- a) F0 is 10 Gb/s per lane FEC ability
- b) F1 is 10 Gb/s per lane FEC requested
- c) F2 is 25G RS-FEC requested
- d) F3 is 25G BASE-R FEC requested

Bits F2 and F3 are used for resolving FEC operation for 25G PHYs, while bits F0 and F1 are used for 10 Gb/s per lane operation. Bits F0 and F1 are not used for 25G PHYs.

73.6.5.1 FEC resolution for 25G PHYs

For 25G PHYs if neither PHY requests FEC operation in bits F2 or F3 then FEC is not enabled.

For 25GBASE-KR and 25GBASE-CR PHYs if either PHY requests RS-FEC then RS-FEC operation is enabled, otherwise if either PHY requests BASE-R FEC then BASE-R operation is enabled.

For 25GBASE-KR-S and 25GBASE-CR-S PHYs, if either PHY requests RS-FEC or BASE-R FEC then BASE-R operation is enabled. This is because 25GBASE-KR-S and 25GBASE-CR-S PHYs do not support RS-FEC operation.

73.6.5.2 FEC resolution for 10 Gb/s per lane PHYs

For 10 Gb/s per lane operation, when the FEC ability bit F0 is set to logical one, it indicates that the PHY has FEC ability (see Clause 74). When the FEC requested F1 bit is set to logical one, it indicates a request to enable FEC on the link.

Since the local device and the link partner may have set the FEC capability bits differently, the priority resolution function is used to enable FEC in the respective PHYs. The FEC function shall be enabled on the link if 10GBASE-KR, 40GBASE-KR4, 40GBASE-CR4, or 100GBASE-CR10 is the HCD technology (see 73.7.6), both devices advertise FEC ability on the F0 bits, and at least one device requests FEC on the F1 bits; otherwise FEC shall not be enabled.

73.6.5.3 FEC control variables

The variable `an_baser_fec_control` indicates that BASE-R FEC operation has been negotiated. If the value is false, then BASE-R FEC has not been negotiated. If the value is true, then BASE-R FEC has been negotiated.

The variable `an_rs_fec_control` indicates that RS-FEC operation has been negotiated. If the value is false, then RS-FEC has not been negotiated. If the value is true, then RS-FEC has been negotiated.

The mapping of these variables to MDIO register bits is defined in Table 73–6.

If `mr_autoneg_enable` (see 73.10.1) is false, the FEC function is controlled by implementation-dependent means.

73.6.6 Pause Ability

Pause (C0:C1) is encoded in bits D11:D10 of the base link codeword. The two-bit Pause is encoded as follows:

- a) C0 is the same as PAUSE as defined in Annex 28B
- b) C1 is the same as ASM_DIR as defined in Annex 28B

The Pause encoding is defined in Clause 28B.2, Table 28B–2. The PAUSE bit indicates that the device is capable of providing the symmetric PAUSE functions as defined in Annex 31B. The ASM_DIR bit indicates that asymmetric PAUSE is supported. The value of the PAUSE bit when the ASM_DIR bit is set indicates the direction the PAUSE frames are supported for flow across the link. Asymmetric PAUSE configuration results in independent enabling of the PAUSE receive and PAUSE transmit functions as defined by Annex 31B. See 28B.3 regarding PAUSE configuration resolution.

73.6.7 Remote Fault

Remote Fault (RF) is encoded in bit D13 of the base link codeword. The default value is logical zero. The Remote Fault bit provides a standard transport mechanism for the transmission of simple fault information. When the RF bit in the AN advertisement register (register 7.16.13) is set to logical one, the RF bit in the transmitted base link codeword is set to logical one. When the RF bit in the received base link codeword is set to logical one, the Remote Fault bit in the AN LP Base Page ability register (register 7.19.13) will be set to logical one, if the management function is present.

73.6.8 Acknowledge

Acknowledge (Ack) is used by the Auto-Negotiation function to indicate that a device has successfully received its link partner's link codeword. The Acknowledge Bit is encoded in bit D14 of link codeword. If no Next Page information is to be sent, this bit shall be set to logical one in the link codeword after the reception of at least three consecutive and consistent DME pages (ignoring the Acknowledge bit value). If Next Page information is to be sent, this bit shall be set to logical one after the device has successfully received at least three consecutive and matching DME pages (ignoring the Acknowledge bit value), and will remain set until the Next Page information has been loaded into the AN XNP transmit register (registers 7.22, 7.23, 7.24). In order to save the current received link codeword, it has to be read from the AN LP XNP ability register (register 7.25, 7.26, 7.27) before the Next Page of transmit information is loaded into the AN XNP transmit register. After the COMPLETE ACKNOWLEDGE state has been entered, the link codeword will be transmitted at least six times.

73.6.9 Next Page

Next Page (NP) is encoded in bit D15 of link codeword. Support of Next Pages is mandatory. If the device does not have any Next Pages to send, the NP bit shall be set to logical zero. If a device wishes to engage in Next Page exchange, it shall set the NP bit to logical one. If a device has no Next Pages to send and its link partner has set the NP bit to logical one, it shall transmit Next Pages with Null message codes and the NP bit set to logical zero while its link partner transmits valid Next Pages. Next page exchanges will occur if either the device or its link partner sets the Next Page bit to logical one. The Next Page function is defined in 73.7.7.

73.6.10 Transmit Switch function

During Auto-Negotiation and prior to entry into the AN_GOOD_CHECK state, the Transmit Switch function shall connect only the DME page generator controlled by the Transmit State Diagram to the MDI.

Upon entry into the AN_GOOD_CHECK state, the Transmit Switch function shall connect the transmit path from a single technology-dependent (highest common denominator) PHY to the MDI.

When a PHY is connected to the MDI through the Transmit Switch function, the signals at the MDI shall conform to all of the PHY's specifications within 20 ms.

73.7 Receive function requirements

The receive function detects the DME page sequence, decodes the information contained within, and stores the data in `rx_link_code_word[48:1]`. The receive function incorporates a receive switch to control connection of the DME page receiver or a PHY to the MDI.

73.7.1 DME page reception

To be able to detect the DME bits, the receiver should have the capability to receive DME signals sent with the electrical specifications of the PHY. The DME transmit signal level and receive sensitivity are specified in 73.5.1.

73.7.2 Receive Switch function

During Auto-Negotiation and prior to entry into the `AN_GOOD_CHECK` state, the Receive Switch function shall connect the DME page receiver to the MDI. For the Parallel Detection function, the Receive Switch function shall also connect the receive path of the 1000BASE-KX and 10GBASE-KX4 PHY to the MDI when those PHYs are present.

Upon entry into the `AN_GOOD_CHECK` state, the Receive Switch function shall connect the receive path from a single technology-dependent (highest common denominator) PHY to the MDI.

73.7.3 Link codeword matching

The Receive function shall generate `ability_match`, `acknowledge_match`, and `consistency_match` variables as defined in Arbitration state diagram Figure 73–11.

73.7.4 Arbitration function requirements

The Arbitration function is described in Figure 73–11 and ensures proper sequencing of the Auto-Negotiation function using the Transmit function and Receive function. The Arbitration function enables the Transmit function to advertise and acknowledge abilities. Upon indication of acknowledgement, the Arbitration function determines the highest common denominator using the priority resolution function and enables the appropriate technology-dependent PHY via the `link_control[x]` variables.

If `mr_autoneg_enable` (see 73.10.1) is false, enabling the desired technology-dependent PHY is controlled by implementation-dependent means.

73.7.4.1 Parallel Detection function

The local device detects a link partner that supports Auto-Negotiation by DME page detection. The Parallel Detection function allows detection of link partners that support 1000BASE-KX, 2.5GBASE-KX, and 10GBASE-KX4 but have disabled Auto-Negotiation, and detection of legacy devices that can interoperate with 1000BASE-KX, 2.5GBASE-KX, and 10GBASE-KX4 devices, but do not provide Clause 73 Auto-Negotiation.

A local device shall provide Parallel Detection for 1000BASE-KX, 2.5GBASE-KX, and 10GBASE-KX4 if it supports those PHYs. Additionally, parallel detection may be used for 10GBASE-CX4. Parallel detection of 10GBASE-CX4 will be indicated by the setting of the Negotiated Port Type to “10GBASE-KX4 or 10GBASE-CX4” in the management register bit 7.48.2. The means to distinguish between 10GBASE-KX4

and 10GBASE-CX4 is implementation dependent. Parallel Detection shall be performed by directing the MDI receive activity to the PHY. This detection may be done in sequence between detection of DME pages and detection of each supported PHY. If at least one of the 1000BASE-KX, 2.5GBASE-KX, or 10GBASE-KX4 establishes link_status=OK, the LINK STATUS CHECK state is entered and the autoneg_wait_timer is started. If exactly one link_status=OK indication is present when the autoneg_wait_timer expires, then Auto-Negotiation shall set link_control=ENABLE for the PHY indicating link_status=OK. If a PHY is enabled, the Arbitration function shall set link_control=DISABLE to all other PHYs and indicate that Auto-Negotiation has completed. On transition to the AN GOOD CHECK state from the LINK STATUS CHECK state, the Parallel Detection function shall set the bit in the AN LP Base Page ability registers (see 45.2.7.7) corresponding to the technology detected by the Parallel Detection function.

If Auto-Negotiation detects link_status=OK from any of the technology-dependent PHYs prior to DME page detection, the autoneg_wait_timer shall start. If more than one technology-dependent PHYs indicate link_status=OK when the autoneg_wait_timer expires, Auto-Negotiation will not allow any data service to be enabled and may signal this as a remote fault to the link partner using the Base Page and will flag this in the local device by setting the Parallel Detection fault bit (45.2.7.2) in the AN Status register.

73.7.5 Renegotiation function

A renegotiation request from any entity, such as a management agent, shall cause the Arbitration function to disable all technology-dependent PHYs and halt any transmit data and link transition activity until the break_link_timer expires. Consequently, the link partner will go into link fail and normal Auto-Negotiation resumes. The local device shall resume Auto-Negotiation after the break_link_timer has expired by issuing DME pages with the Base Page valid in tx_link_code_word[48:1]. Once Auto-Negotiation has completed, renegotiation will take place if the Highest Common Denominator technology that receives link_control=ENABLE returns link_status=FAIL. To allow the PHY an opportunity to determine link integrity using its own link integrity test function, the link_fail_inhibit_timer qualifies the link_status=FAIL indication such that renegotiation takes place if the link_fail_inhibit_timer has expired and the PHY still indicates link_status=FAIL.

73.7.6 Priority Resolution function

Since a local device and a link partner may have multiple common abilities, a mechanism to resolve which mode to configure is required. The mechanism used by Auto-Negotiation is a Priority Resolution function that predefines the hierarchy of supported technologies. The single PHY enabled to connect to the MDI by Auto-Negotiation shall be the technology corresponding to the bit in the Technology Ability Field common

to the local device and link partner that has the highest priority as defined in Table 73–5 (listed from highest priority to lowest priority).

Table 73–5—Priority Resolution

Priority	Technology	Capability
1	200GBASE-KR4 or 200GBASE-CR4	200 Gb/s 4 lane, highest priority
2	100GBASE-KR2 or 100GBASE-CR2	100 Gb/s 2 lane
3	100GBASE-CR4	100 Gb/s 4 lane
4	100GBASE-KR4	100 Gb/s 4 lane
5	100GBASE-KP4	100 Gb/s 4 lane
6	100GBASE-CR10	100 Gb/s 10 lane
7	50GBASE-KR or 50GBASE-CR	50 Gb/s 1 lane
8	40GBASE-CR4	40 Gb/s 4 lane
9	40GBASE-KR4	40 Gb/s 4 lane
10	25GBASE-KR or 25GBASE-CR	25 Gb/s 1 lane
11	25GBASE-KR-S or 25GBASE-CR-S	25 Gb/s 1 lane, short reach
12	10GBASE-KR	10 Gb/s 1 lane
13	10GBASE-KX4	10 Gb/s 4 lane
14	5GBASE-KR	5 Gb/s 1 lane
15	2.5GBASE-KX	2.5 Gb/s 1 lane
16	1000BASE-KX	1 Gb/s 1 lane, lowest priority

The common technology is referred to as the highest common denominator, or HCD, technology. If the local device receives a Technology Ability Field with a bit set that is reserved, the local device shall ignore that bit for priority resolution. Determination of the HCD technology occurs on entrance to the AN GOOD CHECK state. In the event that a technology is chosen through the parallel detection function, that technology shall be considered the highest common denominator (HCD) technology. In the event that there is no common technology, HCD shall have a value of “NULL”, indicating that no PHY receives link_control=ENABLE and link_status[HCD]=FAIL.

NOTE—If both local device and link partner are Backplane Ethernet compliant PHYs, then both ends use abilities exchanged through Clause 73 Auto-Negotiation function. If the Link partner is a legacy device (or has disabled Auto-Negotiation) as indicated by the parallel detect function, then the peer 1 Gb/s devices can opt to use abilities exchanged through Clause 37. This will ensure there are no interoperability issues when connected to a Backplane Ethernet PHY.

73.7.7 Next Page function

The Next Page function uses the Auto-Negotiation arbitration mechanisms to allow exchange of Next Pages of information, which may follow the transmission and acknowledgment procedures used for the base link codeword. The Next Page has both Message code field and Unformatted code fields.

A dual acknowledgment system is used. Acknowledge (Ack) is used to acknowledge receipt of the information; Acknowledge 2 (Ack2) is used to indicate that the receiver is able to act on the information (or perform the task) defined in the message.

The Toggle bit is used to ensure proper synchronization between the local device and the link partner.

Next page exchange occurs after the base link codewords have been exchanged if either end of the link segment set the Next Page bit to logical one indicating that it had at least one Next Page to send. Next page exchange consists of using the normal Auto-Negotiation arbitration process to send Next Page messages.

The Next Page contains two message encodings. The message encodings are defined as follows: message code, which contain predefined 11-bit codes, and unformatted code contains 32 bit codes. Multiple Next Pages with appropriate message codes and unformatted codes can be transmitted to send extended messages. Each series of Next Pages shall have a Message code that defines how the Unformatted codes will be interpreted. Any number of Next Pages may be sent in any order; however, it is recommended that the total number of Next Pages sent be kept small to minimize the link startup time.

Next page transmission ends when both ends of a link segment set their Next Page bits to logical zero, indicating that neither has anything additional to transmit. It is possible for one device to have more pages to transmit than the other device. Once a device has completed transmission of its Next Page information, it shall transmit Next Pages with Null message codes and the NP bit set to logical zero while its link partner continues to transmit valid Next Pages. An Auto-Negotiation able device shall recognize reception of Message Pages with Null message codes as the end of its link partner's Next Page information.

73.7.7.1 Next page encodings

The Next Page shall use the encoding shown in Figure 73–7 and Figure 73–8 for the NP, Ack, MP, Ack2, and T bits. These bits shall function as specified in 28.2.3.4. There are two types of Next Page encodings—message and unformatted. For message Next Pages, the MP bit shall be set to logical one, the 11-bit field D[10:0] shall be encoded as a Message Code Field and D[47:16] shall be encoded as Unformatted Code Field. For Unformatted Next Pages, the MP bit shall be set to logical zero; D[10:0] and D[47:16] shall be encoded as the Unformatted Code Field.

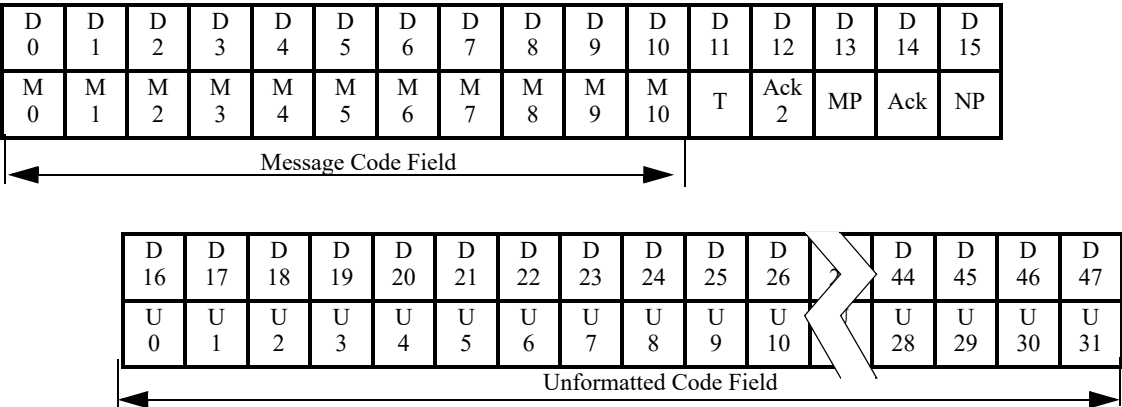


Figure 73–7—Message Next Page

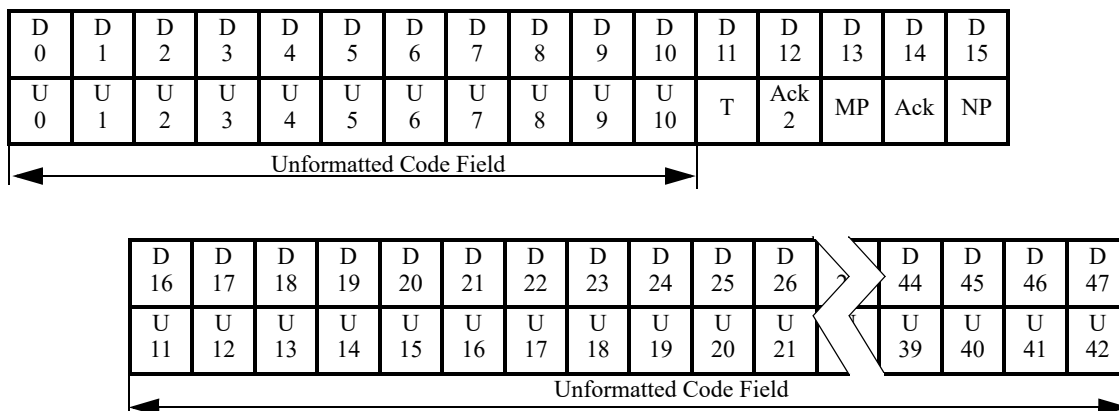


Figure 73–8—Unformatted Next Page

73.7.7.2 Use of Next Pages

Next page exchange will commence after the Base Page exchange if either device requests it by setting the NP bit to logical one.

Next page exchange shall continue until neither device on a link has more pages to transmit as indicated by the NP bit. A Next Page with a Null Message Code Field value shall be sent if the device has no other information to transmit.

A message code can carry either a specific message or information that defines how the corresponding unformatted codes should be interpreted.

73.8 Management register requirements

The management interface is used to communicate Auto-Negotiation information to the management entity. MMD7 of the Clause 45 Management Data Input/Output (MDIO) interface shall be provided as the logical interface to access the device registers for Auto-Negotiation and other management purposes. The Clause 45 MDIO electrical interface is optional. Where no physical embodiment of the MDIO exists, provision of an equivalent mechanism to access the registers is recommended. Table 73–6 provides the mapping of Backplane Ethernet Auto-Negotiation variables to management registers.

**Table 73–6—Backplane Ethernet Auto-Negotiation
variable to MDIO register mapping**

Variable	Description
mr_adv_ability[48:1]	{7.18.15:0, 7.17.15:0, 7.16.15:0} AN advertisement registers
mr_autoneg_complete	7.1.5 Auto-Negotiation Complete
mr_autoneg_enable	7.0.12 Auto-Negotiation Enable
mr_lp_adv_ability[48:1]	For Base Page: {7.21.15:0, 7.20.15:0, 7.19.15:0} AN LP Base Page ability registers For Next Page(s): {7.27.15:0, 7.26.15:0, 7.25.15:0} AN LP XNP ability registers
mr_lp_autoneg_able	7.1.0 LP Auto-Negotiation Able
mr_main_reset	7.0.15 Reset
mr_next_page_loaded	Set on write to AN XNP Transmit registers; cleared by Arbitration state diagram
mr_np_tx[48:1]	{7.24.15:0, 7.23.15:0, 7.22.15:0} AN XNP Transmit registers
mr_page_rx	7.1.6 Page Received
mr_parallel_detection_fault	7.1.9 Parallel detection Fault
mr_restart_negotiation	7.0.9 Auto-Negotiation Restart
set to 1	7.1.3 Auto-Negotiation Ability
link_control_[x]	{7.48.15:8, 7.48.6:5, 7.48.3:1, 7.49.2:0} Negotiated Port Type
an_baser_fec_control	7.48.4 BASE-R FEC negotiated
an_rs_fec_control	7.48.7 RS-FEC negotiated

73.9 Technology-Dependent interface

The Technology-Dependent interface is the communication mechanism between each technology's PCS and the Auto-Negotiation function. Auto-Negotiation can support multiple technologies, all of which need not be implemented in a given device. Each of these technologies may utilize its own technology-dependent link integrity test function.

73.9.1 AN_LINK.indication

This primitive is generated by the PCS to indicate the status of the underlying medium. The purpose of this primitive is to give the Auto-Negotiation function a means of determining the validity of received code elements.

73.9.1.1 Semantics of the service primitive

AN_LINK.indication(link_status)

The link_status parameter shall assume one of two values: OK or FAIL, indicating whether the underlying receive channel is intact and enabled (OK) or not intact (FAIL).

73.9.1.2 When generated

A technology-dependent PCS generates this primitive to indicate a change in the value of link_status.

73.9.1.3 Effect of receipt

The effect of receipt of this primitive shall be governed by the state diagram of Figure 73–11.

73.10 State diagrams and variable definitions

The notation used in state diagrams follows the conventions in Clause 28. Variables in a state diagram with default values evaluate to the variable default in each state where the variable value is not explicitly set.

Auto-Negotiation shall implement the Transmit state diagram, Receive state diagram and Arbitration state diagram. Additional requirements to these state diagrams are made in the respective functional requirements sections. Options to these state diagrams clearly stated as such in the functional requirements sections or state diagrams shall be allowed. In the case of any ambiguity between stated requirements and the state diagrams, the state diagrams shall take precedence.

73.10.1 State diagram variables

A variable with “_x” appended to the end of the variable name indicates a variable or set of variables as defined by “x”. “x” may be as follows:

all;	represents all specific technology-dependent PMDs supported in the local device.
1GKX;	represents the 1000BASE-KX PMD.
2.5GKX;	represents the 2.5GBASE-KX PMD.
5GKR;	represents the 5GBASE-KR PMD.
10GKR;	represents the 10GBASE-KR PMD.
10GKX4;	represents the 10GBASE-KX4 or 10GBASE-CX4 PMD.
25GR;	represents the 25GBASE-KR, 25GBASE-KR-S, 25GBASE-CR, or 25GBASE-CR-S PMD.
40GKR4;	represents the 40GBASE-KR4 PMD.
40GCR4;	represents the 40GBASE-CR4 PMD.
50GR;	represents the 50GBASE-KR or 50GBASE-CR PMD.
100GCR10;	represents the 100GBASE-CR10 PMD.
100GKP4;	represents the 100GBASE-KP4 PMD.
100GKR4;	represents the 100GBASE-KR4 PMD.
100GCR4;	represents the 100GBASE-CR4 PMD.
100GR2;	represents the 100GBASE-KR2 or 100GBASE-CR2 PMD.
200GR4;	represents the 200GBASE-KR4 or 200GBASE-CR4 PMD.
HCD;	represents the single technology-dependent PMD chosen by Auto-Negotiation as the highest common denominator technology through the Priority Resolution or

- parallel detection function.
- notHCD; represents all technology-dependent PMDs not chosen by Auto-Negotiation as the highest common denominator technology through the Priority Resolution or parallel detection function.
- PD; represents all of the following that are present: 1000BASE-KX PMD, 2.5GBASE-KX PMD, and 10GBASE-KX4 (or 10GBASE-CX4) PMD.

Variables with [48:1] appended to the end of the variable name indicate arrays that can be directly mapped to 48-bit registers. For these variables, “[x]” indexes an element or set of elements in the array, where “[x]” may be as follows:

- a) Any integer
- b) Any range of integers
- c) Any variable that takes on integer values
- d) NP; represents the index of the Next Page bit
- e) ACK; represents the index of the Acknowledge bit
- f) RF; represents the index of the Remote Fault bit

Variables of the form “mr_x”, where x is a label, comprise a management interface that is intended to be connected to the Management function. However, an implementation-specific management interface may provide the control and status function of these bits.

ability_match

Indicates that three consecutive link codewords match, ignoring the Acknowledge bit. Three consecutive words are any three words received one after the other, regardless of whether the word has already been used in a word-match comparison or not.

Values: false; three matching consecutive link codewords have not been received, ignoring the Acknowledge bit (default).
true; three matching consecutive link codewords have been received, ignoring the Acknowledge bit.

NOTE—This variable is set by this variable definition; it is not set explicitly in the state diagrams.

ability_match_word [48:1]

A 48-bit array that is loaded upon transition to Acknowledge Detect state with the value of the link codeword that caused ability_match = true for that transition. For each element in the array transmitted.

Values: zero; data bit is logical zero.
one; data bit is logical one.

NOTE—This variable is set by this variable definition; it is not set explicitly in the state diagrams.

ack_finished

Status indicating that the final remaining_ack_cnt link codewords with the Ack bit set have been transmitted.

Values: false; more link codewords with the Ack bit set to logical one have to be transmitted.
true; all remaining link codewords with the Ack bit set to logical one have been transmitted.

ack_nonce_match

Indicates whether the echoed nonce received from the link partner matches the transmitted nonce

field sent by the local device. The echoed nonce value from the DME page that caused `acknowledge_match` to be set is used for this test.

Values `false`; link partner echoed nonce does not equal local device transmitted nonce.
 `true`; link partner echoed nonce equals local device transmitted nonce.

`acknowledge_match`

Indicates that three consecutive link codewords match and have the Acknowledge bit set. Three consecutive words are any three words received one after the other, regardless of whether the word has already been used in a word match comparison or not.

Values: `false`; three matching and consecutive link codewords have not been received with the Acknowledge bit set (default).
 `true`; three matching and consecutive link codewords have been received with the Acknowledge bit set.

NOTE—This variable is set by this variable definition; it is not set explicitly in the state diagrams.

`an_link_good`

Indicates that Auto-Negotiation has completed.

Values: `false`; negotiation is in progress (default).
 `true`; negotiation is complete, forcing the Transmit and Receive functions to IDLE.

`an_receive_idle`

Indicates that the Receive state diagram is in the IDLE or DELIMITER DETECT state.

Values: `false`; the Receive state diagram is not in the IDLE or DELIMITER DETECT state (default).
 `true`; the Receive state diagram is in the IDLE or DELIMITER DETECT state.

`base_page`

Status indicating that the page currently being transmitted by Auto-Negotiation is the initial link codeword encoding used to communicate the device's abilities.

Values: `false`; a page other than base link codeword is being transmitted.
 `true`; the base link codeword is being transmitted.

`code_sel`

A Boolean random or pseudo-random value uniformly distributed. A new value is generated each time the variable `code_sel` is used.

Values: `zero`; a zero has been assigned.
 `one`; a one has been assigned.

`complete_ack`

Controls the counting of transmitted link codewords that have their Acknowledge bit set.

Values: `false`; transmitted link codewords with the Acknowledge bit set are not counted (default).
 `true`; transmitted link codewords with the Acknowledge bit set are counted.

`consistency_match`

Indicates that the `ability_match_word` same as the link codeword that caused `acknowledge_match` to be set.

Values: `false`; the link codeword that caused `ability_match` to be set is not the same as the link codeword that caused `acknowledge_match` to be set, ignoring the Acknowledge bit value and the echoed nonce value.

true; the link codeword that caused ability_match to be set is the same as the link codeword that caused acknowledge_match to be set, ignoring the Acknowledge bit value and the echoed nonce value.

NOTE—This variable is set by this variable definition; it is not set explicitly in the state diagrams.

detect_mv_pair

Status indicating that the receiver has detected the pair of Manchester violations forming a Manchester Violation delimiter—a sequence of three consecutive transitions with $12.8 \text{ ns} \pm 200 \text{ ps}$ between each pair of transitions.

Values: false; set to false after any Receive State Diagram state transition (default).
true; Manchester violation pair has been detected.

detect_transition

Status indicating that the receiver has detected a transition.

Values: false; set to false after any Receive State Diagram state transition (default).
true; set to true when a transition is received.

incompatible_link

Parameter used following Priority Resolution to indicate the resolved link is incompatible with the local device settings. A device's ability to set this variable to true is optional.

Values: false; A compatible link exists between the local device and link partner (default).
true; Optional indication that Priority Resolution has determined no highest common denominator exists following the most recent negotiation.

NOTE—This variable is set by this variable definition; it is not set explicitly in the state diagrams.

link_control_[x]

Controls the connection of each PMD to the MDI. When all PMD transmitters are isolated from the MDI, the AN transmitter is connected to the MDI.

Values: DISABLE; isolates the PMD from the MDI.
SCAN_FOR_CARRIER; connects the PMD receiver to the MDI and isolates the PMD transmitter from the link.
ENABLE; connects the PMD (both transmit and receive) to the MDI.

link_status_[x]

This variable is defined in 73.9.1.

mr_autoneg_complete

Status indicating whether Auto-Negotiation has completed or not.

Values: false; Auto-Negotiation has not completed.
true; Auto-Negotiation has completed.

mr_autoneg_enable

Controls the enabling and disabling of the Auto-Negotiation function.

Values: false; Auto-Negotiation is disabled.
true; Auto-Negotiation is enabled.

mr_adv_ability[48:1]

A 48-bit array that contains the Advertised Abilities link codeword.
For each element within the array:

Values: zero; data bit is logical zero.

one; data bit is logical one.

mr_lp_adv_ability[48:1]

A 48-bit array that contains the link partner's Advertised Abilities link codeword.

For each element within the array:

Values: zero; data bit is logical zero.

one; data bit is logical one.

mr_lp_autoneg_able

Status indicating whether the link partner supports Auto-Negotiation.

Values: false; the link partner does not support Auto-Negotiation.

true; the link partner supports Auto-Negotiation.

mr_main_reset

Controls the resetting of the Auto-Negotiation state diagrams.

Values: false; do not reset the Auto-Negotiation state diagrams.

true; reset the Auto-Negotiation state diagrams.

mr_next_page_loaded

Status indicating whether a new page has been loaded into the AN XNP transmit register (45.2.7.8).

Values: false; a New Page has not been loaded.

true; a New Page has been loaded.

mr_np_tx[48:1]

A 48-bit array that contains the new Next Page to transmit.

For each element within the array:

Values: zero; data bit is logical zero.

one; data bit is logical one.

mr_page_rx

Status indicating whether a New Page has been received. A New Page has been successfully received when `acknowledge_match=true` and `consistency_match=true` and the link codeword has been written to `mr_lp_adv_ability[48:1]`.

Values: false; a New Page has not been received.

true; a New Page has been received.

mr_parallel_detection_fault

Error condition indicating that while performing parallel detection, either `an_receive_idle = false`, or zero or more than one of the following indications were present when the `autoneg_wait_timer` expired. This signal is cleared on read of the AN status register (register 7.1).

1) `link_status_[1GKX] = OK`

2) `link_status_[10GKX4] = OK`

Values: false; Exactly one of the above two indications was true when the `autoneg_wait_timer` expired, and `an_receive_idle = true`.

true; either zero or more than one of the above two indications was true when the `autoneg_wait_timer` expired, or `an_receive_idle = false`.

mr_restart_negotiation

Controls the entrance to the TRANSMIT DISABLE state to break the link before Auto-Negotiation is allowed to renegotiate via management control.

Values: false; renegotiation is not taking place.
true; renegotiation is started.

nonce_match

Indicates whether the transmitted nonce received from the link partner matches the transmitted nonce field sent by the local device.

Values false; link partner transmitted nonce does not equal local device transmitted nonce.
true; link partner transmitted nonce equals local device transmitted nonce.

np_rx

Flag to hold the value of rx_link_code_word[NP] upon entry to the COMPLETE ACKNOWLEDGE state. This value is associated with the value of rx_link_code_word[NP] when acknowledge_match was last set.

Values zero; local device np_rx bit equals a logical zero.
one; local device np_rx bit equals a logical one.

power_on

Condition that is true until such time as the power supply for the device that contains the Auto-Negotiation state diagrams has reached the operating region or the device has low-power mode set via MMD control register bit 1.120.12.

Values: false; the device is completely powered (default).
true; the device has not been completely powered.

pulse_too_long

Indicates that the receiver has detected successive transitions spaced too far apart for a valid DME page. Transitions separated by more than 20 ns shall cause this indication to be true. Valid Manchester violation delimiters shall not cause this indication to be true.

Values: false; excessively long pulses have not been detected.
true; excessively long pulses have been detected.

pulse_too_short

Indicates that the receiver has detected successive transitions spaced too closely for a valid DME page. Transitions separated by less than 1.6 ns shall cause this indication to be true. Valid Manchester transitions shall not cause this indication to be true.

Values: false; excessively short pulses have not been detected.
true; excessively short pulses have been detected.

rx_link_code_word[48:1]

A 48-bit array that contains the data bits to be received from a DME page.
For each element within the array:

Values: zero; data bit is a logical zero.
one; data bit is a logical one.

rx_nonce[4:0]

A 5-bit array that contains the transmitted nonce received from the DME page that caused ability_match=true.
For each element within the array:

Values: zero; data bit is a logical zero.
one; data bit is a logical one.

single_link_ready

Status indicating that an_receive_idle = true and only one the of the following indications is being

received:

- 1) link_status_[1GKX] = OK
- 2) link_status_[2.5GKX] = OK
- 3) link_status_[5GKR] = OK
- 4) link_status_[10GKX4] = OK
- 5) link_status_[10GKR] = OK
- 6) link_status_[25GR] = OK
- 7) link_status_[40GKR4] = OK
- 8) link_status_[40GCR4] = OK
- 9) link_status_[50GR] = OK
- 10) link_status_[100GCR10] = OK
- 11) link_status_[100GKP4] = OK
- 12) link_status_[100GKR4] = OK
- 13) link_status_[100GCR4] = OK
- 14) link_status_[100GR2] = OK
- 15) link_status_[200GR4] = OK

Values: false; either zero or more than one of the above indications are true or an_receive_idle = false.

true; Exactly one of the above indications is true and an_receive_idle = true.

NOTE—This variable is set by this variable definition; it is not set explicitly in the state diagrams.

TD_AUTONEG

Controls the signal sent by Auto-Negotiation on the TD_AUTONEG circuit.

Values: disable; transmission of Auto-Negotiation signals is disabled
idle; Auto-Negotiation maintains the current signal level on the MDI.
mv_delimiter; Auto-Negotiation causes the transmission of the Manchester violation delimiter on the MDI.
transition; Auto-Negotiation causes a transition in the level on the MDI.

toggle_rx

Flag to keep track of the state of the link partner's Toggle bit.

Values: zero; link partner's Toggle bit equals logical zero.
one; link partner's Toggle bit equals logical one.

toggle_tx

Flag to keep track of the state of the local device's Toggle bit.

Values: zero; local device's Toggle bit equals logical zero.
one; local device's Toggle bit equals logical one.

transmit_ability

Controls the transmission of the link codeword containing tx_link_code_word[48:1].

Values: false; any transmission of tx_link_code_word[48:1] is halted (default).
true; the transmit state diagram begins sending tx_link_code_word[48:1].

transmit_ack

Controls the setting of the Acknowledge bit in the tx_link_code_word[48:1] to be transmitted.

Values: false; sets the Acknowledge bit in the transmitted tx_link_code_word[48:1] to a logical zero (default).

true; sets the Acknowledge bit in the transmitted tx_link_code_word[48:1] to a logical one.

transmit_disable

Controls the transmission of tx_link_code_word[48:1].

Values: false; tx_link_code_word[48:1] transmission is allowed (default).
true; tx_link_code_word[48:1] transmission is halted.

transmit_mv_done

Status indicating that the transmission of the Manchester violation delimiter has been completed.

Values: false; transmission of the Manchester violation is in progress.
true; transmission of the Manchester violation has been completed.

tx_link_code_word[49:1]

A 49-bit array that contains the data bits to be transmitted in an DME page.

tx_link_code_word[48:1] contains the Auto-Negotiation page to be transmitted.

tx_link_code_word[49] contains the pseudo-random bit. This array may be loaded from mr_adv_ability or mr_np_tx.

For each element within the array:

Values: zero; data bit is logical zero.
one; data bit is logical one.

73.10.2 State diagram timers

All timers operate in the manner described in 14.2.3.2.

autoneg_wait_timer

Timer for the amount of time to wait before evaluating the number of link integrity test functions with link_status=OK asserted. The autoneg_wait_timer shall expire 25 ms to 50 ms from the assertion of link_status=OK from the PCS.

break_link_timer

Timer for the amount of time to wait in order to assure that the link partner enters a Link Fail state. The timer shall expire 60 ms to 75 ms after being started.

clock_detect_min_timer

Timer for the minimum time between detection of differential Manchester clock transitions. The clock_detect_min_timer shall expire 4.8 ns to 6.2 ns after being started or restarted.

clock_detect_max_timer

Timer for the maximum time between detection of differential Manchester clock transitions. The clock_detect_max_timer shall expire 6.6 ns to 8.0 ns after being started or restarted.

data_detect_max_timer

Timer for the maximum time between a clock transition and the following data transition. This timer is used in conjunction with the data_detect_min_timer to detect whether the data bit between two clock transitions is a logical zero or a logical one. The data_detect_max_timer shall expire 3.4 ns to 4.8 ns from the last clock transition.

data_detect_min_timer

Timer for the minimum time between a clock transition and the following data transition. This timer is used in conjunction with the data_detect_max_timer to detect whether the data bit between two clock transitions is a logical zero or a logical one. The data_detect_min_timer shall expire 1.6 ns to 3.0 ns from the last clock transition.

interval_timer

Timer for the separation of a transmitted clock pulse from a data bit. The `interval_timer` shall expire $3.2 \text{ ns} \pm 0.01\%$ from each clock pulse and data bit.

link_fail_inhibit_timer

Timer for qualifying a `link_status=FAIL` indication or a `link_status=OK` indication when a specific technology link is first being established. A link will only be considered “failed” if the `link_fail_inhibit_timer` has expired and the link has still not gone into the `link_status=OK` state. The `link_fail_inhibit_timer` shall expire within the timer values given in Table 73–7 after entering the AN LINK GOOD CHECK state.

NOTE—The `link_fail_inhibit_timer` expiration value has to be greater than the time required for the link partner to complete Auto-Negotiation after the local device has completed Auto-Negotiation plus the time required for the specific technology to enter the `link_status=OK` state.

page_test_max_timer

Timer for the maximum time between detection of Manchester violation delimiters. This timer is used in conjunction with the `page_test_min_timer` to detect whether the link partner is transmitting DME pages. The `page_test_max_timer` shall expire 350 ns to 375 ns after being started or restarted.

page_test_min_timer

Timer for the minimum time between detection of Manchester violation delimiters. This timer is used in conjunction with the `page_test_max_timer` to detect whether the link partner is transmitting DME pages. The `page_test_min_timer` shall expire 305 ns 330 ns after being started or restarted.

Table 73–7—Timer min/max value summary

Parameter	Min	Value and tolerance	Max	Units
<code>autoneg_wait_timer</code>	25		50	ms
<code>break_link_timer</code>	60		75	ms
<code>clock_detect_min_timer</code>	4.8		6.2	ns
<code>clock_detect_max_timer</code>	6.6		8.0	ns
<code>data_detect_min_timer</code>	1.6		3.0	ns
<code>data_detect_max_timer</code>	3.4		4.8	ns
<code>interval_timer</code>		$3.2 \pm 0.01\%$		ns
<code>link_fail_inhibit_timer</code> (when the link is 50GBASE-KR, 50GBASE-CR, 100GBASE-KR2, 100GBASE-CR2, 200GBASE-KR4 or 200GBASE-CR4)	3.1		3.2	s
<code>link_fail_inhibit_timer</code> (when the link is 10GBASE-KR, 25GBASE-KR, 25GBASE-KR-S, 25GBASE-CR, 25GBASE-CR-S, 40GBASE-KR4, 40GBASE-CR4, 100GBASE-KR4, 100GBASE-KP4, 100GBASE-CR4, or 100GBASE-CR10)	500		510	ms
<code>link_fail_inhibit_timer</code> (when the link is 1000BASE-KX, 2.5GBASE-KX, 5GBASE-KR, or 10GBASE-KX4)	40		50	ms
<code>page_test_min_timer</code>	305		330	ns
<code>page_test_max_timer</code>	350		375	ns

73.10.3 State diagram counters

remaining_ack_cnt

A counter that may take on integer values from 0 to 8. The number of additional link codewords with the Acknowledge Bit set to logical one to be sent to ensure that the link partner receives the acknowledgment.

Values: not_done; positive integers between 0 and 5 inclusive.
done; positive integers 6 to 8 inclusive (default).
init; counter is reset to zero.

rx_bit_cnt

A counter that may take on integer values from 0 to 49. This counter is used to keep a count of data bits received from a DME page and to ensure that when erroneous extra transitions are received, the first 48 bits are kept while the rest are ignored. When this variable reaches 49, enough data bits have been received. This counter does not increment beyond 49 and does not return to 0 until it is reinitialized.

Values: not_done; 0 to 48 inclusive.
done; 49
init; counter is reset to zero.

tx_bit_cnt

A counter that may take on integer values from 1 to 50. This counter is used to keep a count of data bits sent within a DME page. When this variable reaches 50, all data bits have been sent.

Values: not_done; 1 to 49 inclusive.
done; 50.
init; counter is initialized to 1.

73.10.4 State diagrams

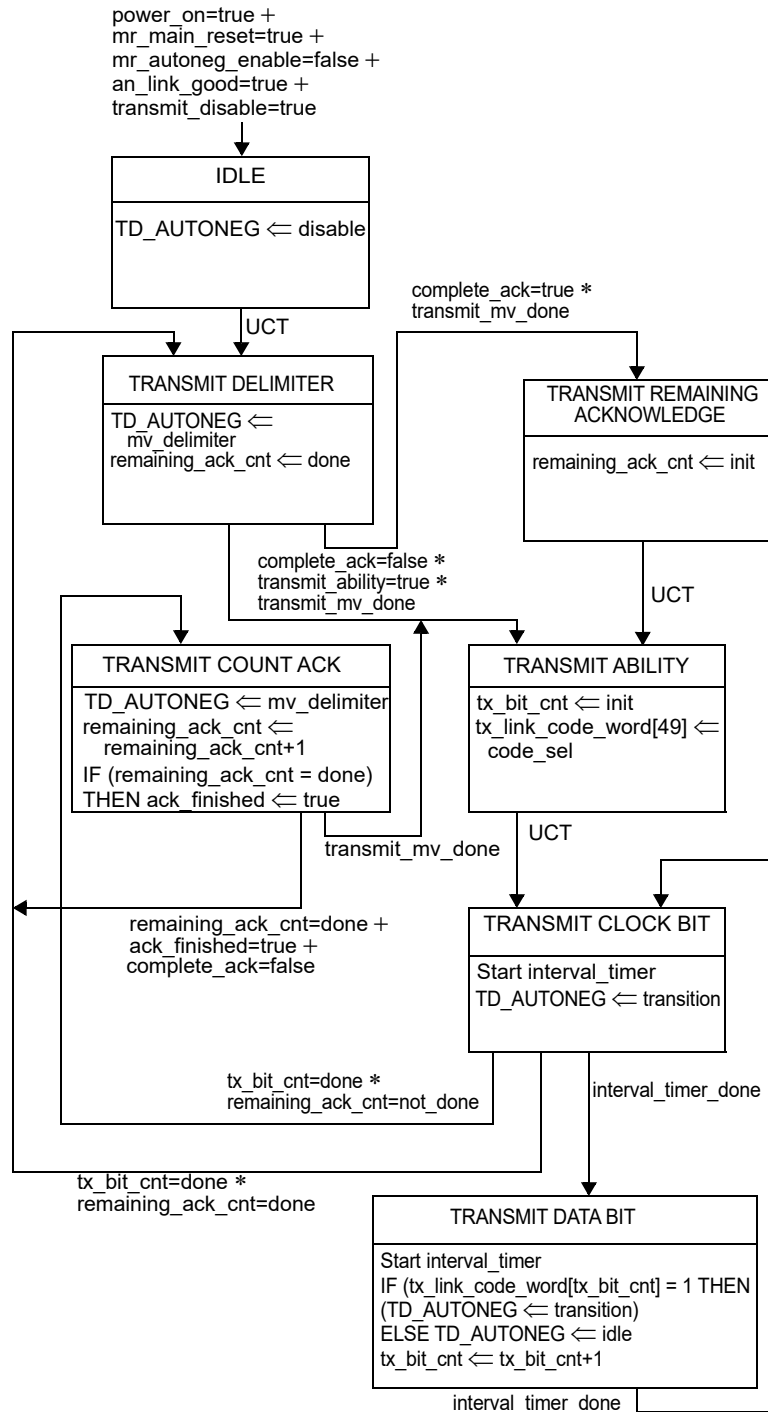


Figure 73–9—Transmit state diagram

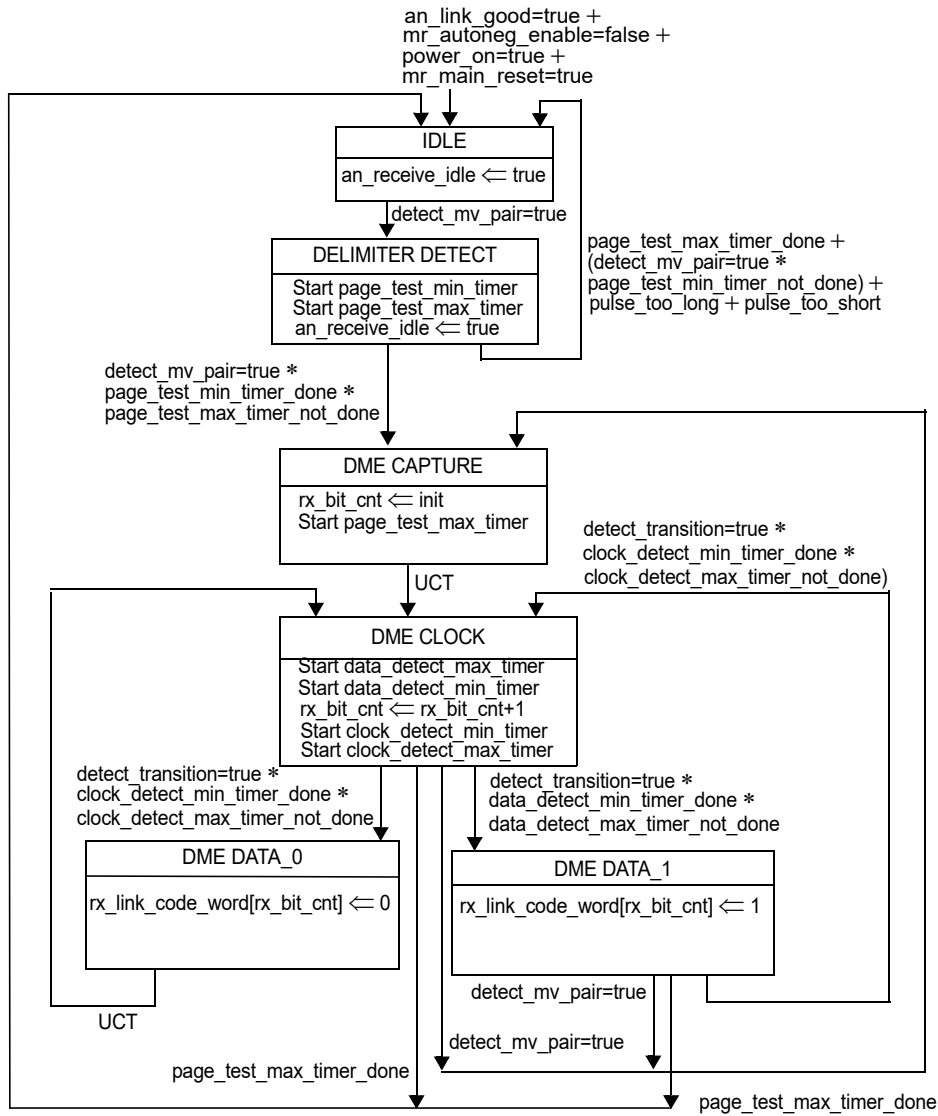


Figure 73–10—Receive state diagram

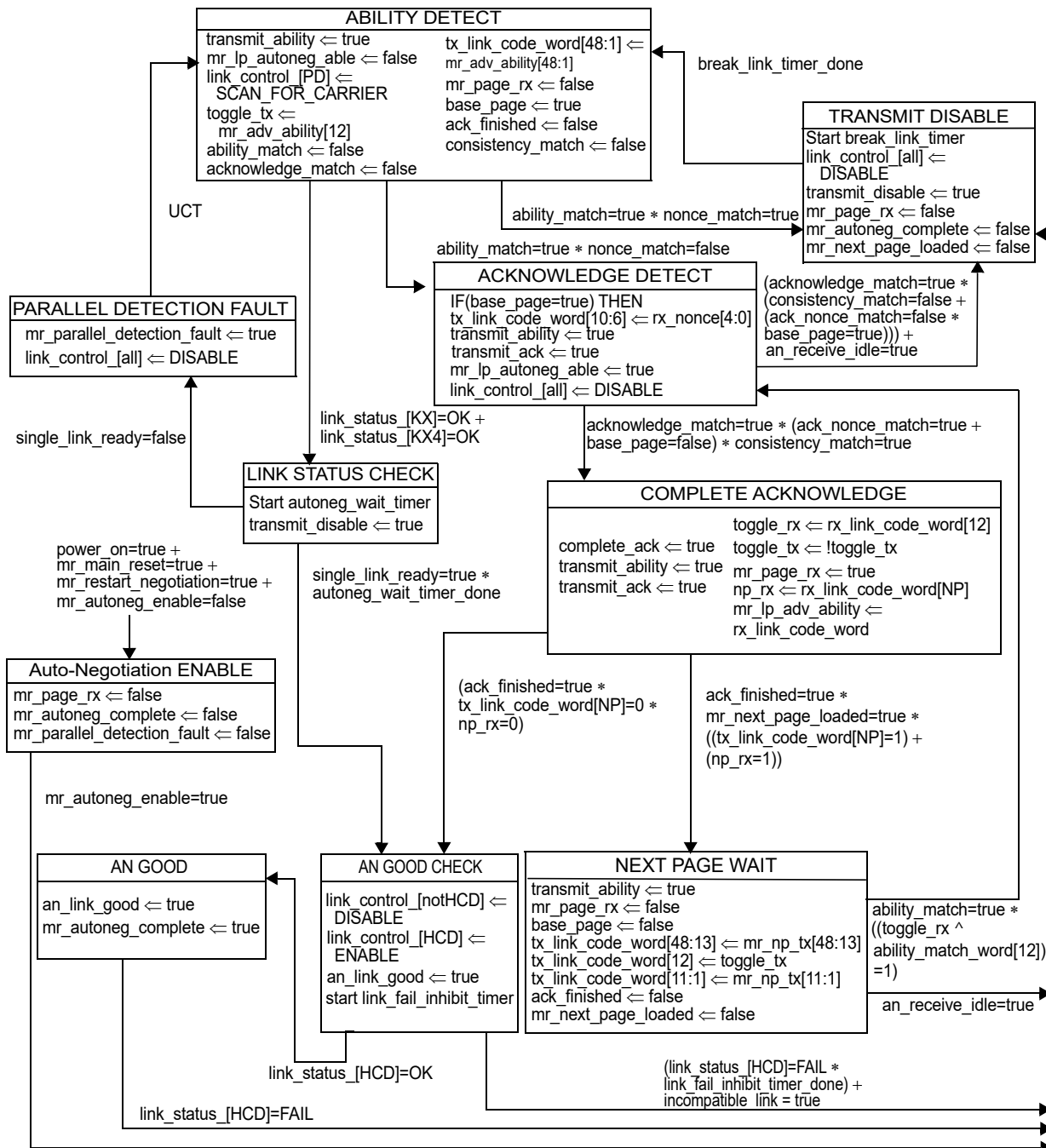


Figure 73–11—Arbitration state diagram

73.11 Protocol implementation conformance statement (PICS) proforma for Clause 73, Auto-Negotiation for backplane and copper cable assembly¹³⁶

73.11.1 Introduction

The supplier of a protocol implementation that is claimed to conform to IEEE Std 802.3, Clause 73, Auto-Negotiation for backplane and copper cable assembly, shall complete the following protocol implementation conformance statement (PICS) proforma. A detailed description of the symbols used in the PICS proforma, along with instructions for completing the PICS proforma, can be found in Clause 21.

73.11.2 Identification

73.11.2.1 Implementation identification

Supplier	
Contact point for inquiries about the PICS	
Implementation Name(s) and Version(s)	
Other information necessary for full identification—e.g., name(s) and version(s) for machines and/or operating systems; System Name(s)	
NOTE 1—Only the first three items are required for all implementations; other information may be completed as appropriate in meeting the requirements for the identification.	
NOTE 2—The terms Name and Version should be interpreted appropriately to correspond with a supplier's terminology (e.g., Type, Series, Model).	

73.11.2.2 Protocol summary

Identification of protocol standard	IEEE Std 802.3-2022, Clause 73, Auto-Negotiation for backplane and copper cable assembly
Identification of amendments and corrigenda to this PICS proforma that have been completed as part of this PICS	
Have any Exception items been required? No [] Yes [] (See Clause 21; the answer Yes means that the implementation does not conform to IEEE Std 802.3-2022.)	
Date of Statement	

¹³⁶Copyright release for PICS proformas: Users of this standard may freely reproduce the PICS proforma in this subclause so that it can be used for its intended purpose and may further publish the completed PICS.

73.11.3 Major capabilities/options

Item	Feature	Subclause	Value/Comment	Status	Support
ANG	Auto-Negotiation	73.1		M	Yes []
PLD	Parallel detection	73.1		M	Yes []
NP	Next page support	73.6.9		M	Yes []

73.11.4 PICS proforma tables for Auto-Negotiation for backplane and copper cable assembly

73.11.4.1 Functional specifications

Item	Feature	Subclause	Value/Comment	Status	Support
FS1	Clause 37 Auto-Negotiation	73.1	Clause 37 Auto-Negotiation to be disabled	O	Yes [] No []
FS2	Device intends to enable Clause 37 Auto-Negotiation after Clause 73 Auto-Negotiation	73.1	Ensure that link partner intends to enable Clause 37 Auto-Negotiation	FS1:M	Yes []
FS3	Advertised abilities for Clause 37 Auto-Negotiation	73.1	Shall match those advertised in Clause 73 Auto-Negotiation	FS1:M	Yes []
FS4	Auto-Negotiation functions	73.3	Auto-Negotiation function shall provide transmit, receive, and arbitration	M	Yes []
FS5	Compliance with state diagrams	73.3	Figure 73–9 through Figure 73–11	M	Yes []
FS6	Interaction with PHYs	73.3	Auto-Negotiation shall interact with technology-dependent PHYs through Technology-Dependent interface.	M	Yes []

73.11.4.2 DME transmission

Item	Feature	Subclause	Value/Comment	Status	Support
DT1	Transmission of DME pages	73.5	DME pages shall not be transmitted when Auto-Negotiation is complete and HCD PHY has been enabled	M	Yes []
DT2	Multi-lane DME pages	73.5.1	Multi-lane DME pages shall be transmitted on Lane 0	M	Yes []
DT3	DME electrical characteristics	73.5.1	Meet requirements of Table 73–1	M	Yes []
DT4	Transitions in odd numbered positions	73.5.2	Remaining 49 odd-numbered positions shall contain a transition	M	Yes []
DT5	Transitions in even numbered positions	73.5.2	Remaining 49 even-numbered positions shall represent data	M	Yes []
DT6	First 48 even numbered positions	73.5.2	First 48 even numbered positions shall carry data of Auto-Negotiation page	M	Yes []
DT7	Pseudo-random bit value	73.5.2	Value shall be derived from source as defined in 48.2.4.2	M	Yes []
DT8	Pseudo-random counter	73.5.2	Counter shall increment once per DME page	M	Yes []
DT9	DME page timing parameters	73.5.3	Meet requirements of Table 73–2	M	Yes []

73.11.4.3 Link codeword encoding

Item	Feature	Subclause	Value/Comment	Status	Support
LE1	Link codeword encoding	73.6	As shown in Figure 73–6	M	Yes []
LE2	First bit transmitted	73.6	D0	M	Yes []
LE3	RF, ACK, NP bits	73.6	As specified in 28.2.1.1	M	Yes []
LE4	Reserved Selector Field	73.6.1	Shall not be transmitted	M	Yes []
LE5	Echoed Nonce Field with Acknowledge set to zero	73.6.2	Contains logical zeros	M	Yes []
LE6	Echoed Nonce Field with Acknowledge set to one	73.6.2	Values received in Transmitted Nonce Field from Link Partner	M	Yes []
LE7	Transmitted Nonce Field	73.6.3	New value generated for each entry into Ability Detect	M	Yes []
LE8	Support of multiple technologies	73.6.4	Shall support all technologies advertised	M	Yes []
LE9	Technology ability reserved fields	73.6.4	Sent as zero and ignored by the receiver	M	Yes []
LE10	FEC capability resolution	73.6.5	Resolve enabling of FEC capability based on F0 and F1 bits	M	Yes []
LE11	Acknowledge with no Next Page	73.6.8	Set to 1 after three DME pages	M	Yes []
LE12	Acknowledge with Next Page	73.6.8	Set to 1 after three DME pages	M	Yes []
LE13	Device has no Next Pages to send	73.6.9	Next Page bit set to 0	M	Yes []
LE14	Device has Next Pages to send	73.6.9	Next Page bit set to 1	M	Yes []
LE15	Transmit switch function upon entry into the AN_GOOD_CHECK state	73.6.10	Connect the transmit path of HCD PHY to the MDI	M	Yes []
LE16	Transmit switch function during Auto-Negotiation and prior to entry into the AN_GOOD_CHECK state	73.6.10	Connect only DME page generator to MDI	M	Yes []
LE17	PHY connection to MDI	73.6.10	Signals at MDI conform to all PHY specifications within 20 ms	M	Yes []

73.11.4.4 Receive function requirements

Item	Feature	Subclause	Value/Comment	Status	Support
RF1	Receive switch function upon entry into the AN_GOOD_CHECK state	73.7.2	Connect the receive path of HCD PHY to the MDI	M	Yes []
RF2	Receive switch function during Auto-Negotiation and prior to entry into the AN_GOOD_CHECK state	73.7.2	Connect DME page receiver to MDI	M	Yes []
RF3	Receive switch function during Auto-Negotiation and prior to entry into the AN_GOOD_CHECK state	73.7.2	Connect receive path of the 1000BASE-KX and 10GBASE-KX4 PHY when present	M	Yes []
RF4	Receive function variables	73.7.3	As defined in Figure 73–11	M	Yes []
RF5	Parallel detection for 1000BASE-KX, 2.5GBASE-KX, and 10GBASE-KX4	73.7.4.1	Device provides parallel detection if it supports those PHYs	M	Yes []
RF6	Parallel detection	73.7.4.1	Direct MDI receive activity to PHYs prior to DME detection	M	Yes []
RF7	Enable one link after parallel detection	73.7.4.1	Enable link if signaling is present	M	Yes []
RF8	Disable all other links after parallel detection	73.7.4.1	Disable all other PHYs	M	Yes []
RF9	Parallel detection register settings	73.7.4.1	Set bit corresponding to technology detected	M	Yes []
RF10	Detection of link_status=OK	73.7.4.1	autoneg_wait_timer starts	M	Yes []
RF11	Renegotiation request	73.7.5	Disable PHYs and halt transmissions for break_link_timer	M	Yes []
RF12	Resumption of Auto-Negotiation	73.7.5	Resume Auto-Negotiation after expiration of break_link_timer	M	Yes []
RF13	Priority resolution	73.7.6	PHY with highest priority connected to MDI	M	Yes []
RF14	Reception of reserved technology ability field bits	73.7.6	Ignore reserved technology ability field bits	M	Yes []
RF15	Priority resolution through parallel detection	73.7.6	PHY chose through parallel detection is HCD	M	Yes []
RF16	Priority resolution with no common technology	73.7.6	HCD takes on value of NULL and link_status=FAIL	M	Yes []

73.11.4.5 Next Page function

Item	Feature	Subclause	Value/Comment	Status	Support
NP1	Message codes	73.7.7	Each series of Next Pages has Message code	M	Yes []
NP2	Next Page transmission while link partner not done	73.7.7	Device transmits Null Message code and sets NP bit to 0	M	Yes []
NP3	Reception of Null message codes	73.7.7	Recognized as end of link partner's Next Pages	M	Yes []
NP4	Next page encoding	73.7.7.1	As shown in Figure 73–7 and Figure 73–8	M	Yes []
NP5	NP, Ack, MP, Ack2, T bits	73.7.7.1	As specified in 28.2.3.4	M	Yes []
NP6	MP bit for message Next Pages	73.7.7.1	Set to logical one	M	Yes []
NP7	Message Code Field in message Next Page	73.7.7.1	Encoded in D[10:0]	M	Yes []
NP8	Unformatted Code Field in message Next Page	73.7.7.1	Encoded in D[47:16]	M	Yes []
NP9	MP bit for Unformatted Next Pages	73.7.7.1	Set to logical zero	M	Yes []
NP10	Unformatted Code Field in Unformatted Next Pages	73.7.7.1	Encoded in D[15:0] and D[47:16]	M	Yes []
NP11	Continuation of Next Page exchange	73.7.7.2	Exchange continues until NP bit is zero on both devices	M	Yes []
NP12	Transmission of Null Message Code field	73.7.7.2	Sent if device has no other information to transmit	M	Yes []

73.11.4.6 Management register requirements

Item	Feature	Subclause	Value/Comment	Status	Support
MR1	MMD 7 of Clause 45 MDIO	73.8	Logical interface for access to device registers	M	Yes []
MR2	Clause 45 electrical interface	73.8	Electrical interface for access to device registers	O	Yes [] No []

73.11.4.7 State diagrams and variable definitions

Item	Feature	Subclause	Value/Comment	Status	Support
SD1	Support of state diagrams	73.10	Transmit, Receive, Arbitration	M	Yes []
SD2	Support of options	73.10	Options are allowed	M	Yes []
SD3	Ambiguity between state diagrams and text	73.10	State diagrams take precedence	M	Yes []
SD4	Pulse too short	73.10.1	Transitions separated by less than 1.6 ns	M	Yes []
SD5	Pulse too short with valid transitions	73.10.1	Valid transitions not to cause this to be true	M	Yes []
SD6	Pulse too long	73.10.1	Transitions separated by more than 20 ns	M	Yes []
SD7	Pulse too long with valid violation delimiters	73.10.1	Valid Manchester violation delimiters not to set this	M	Yes []
SD8	autoneg_wait_timer	73.10.2	25 ms to 50 ms	M	Yes []
SD9	break_link_timer	73.10.2	60 ms to 75 ms	M	Yes []
SD10	clock_detect_min_timer	73.10.2	4.8 ns to 6.2 ns	M	Yes []
SD11	clock_detect_max_timer	73.10.2	6.6 ns to 8.0 ns	M	Yes []
SD12	data_detect_max_timer	73.10.2	4.0 ns to 4.8 ns	M	Yes []
SD13	data_detect_min_timer	73.10.2	1.6 ns to 2.4 ns	M	Yes []
SD14	interval_timer	73.10.2	3.2 ns $\pm$ 0.01%	M	Yes []
SD15	link_fail_inhibit_timer	73.10.2	The values in Table 73–7	M	Yes []
SD16	page_test_max_timer	73.10.2	350 ns to 375 ns	M	Yes []
SD17	page_test_min_timer	73.10.2	305 ns to 330 ns	M	Yes []

73.11.4.8 Service primitives

Item	Feature	Subclause	Value/Comment	Status	Support
SP1	link_status parameter	73.9.1.1	OK, FAIL	M	Yes []
SP2	Generation of link_status primitive	73.9.1.2	Generated by technology dependent PCS	M	Yes []
SP3	Receipt of link_status primitive	73.9.1.3	Governed by Figure 73–10	M	Yes []

73.11.4.9 Auto-Negotiation annexes

Item	Feature	Subclause	Value/Comment	Status	Support
AN1	Null Message code	73A.1	Transmitted during Next Page exchange when local device has no information to transmit and link partner has additional pages to transmit	M	Yes []
AN2	OUI message code	73A.2	0000 0000 0101	M	Yes []
AN3	OUI first user code	73A.2	OUI or CID (bits 23:13)	M	Yes []
AN4	OUI second user code	73A.2	OUI or CID (bits 12:2)	M	Yes []
AN5	OUI third user code	73A.2	OUI or CID (bits 1:0)	M	Yes []
AN6	OUI fourth user code	73A.2	User-defined code value	M	Yes []
AN7	AN device identifier Message code	73A.3	0000 0000 0110	M	Yes []
AN8	AN device identifier first user code	73A.3	AN device identifier (7.2.15:5)	M	Yes []
AN9	AN device identifier second user code	73A.3	AN device identifier (7.2.4:0 to 7.3.15:10)	M	Yes []
AN10	AN device identifier third user code	73A.3	AN device identifier (7.3.9:0)	M	Yes []
AN11	AN device identifier third user code bit 0	73A.3	User-defined code value	M	Yes []
AN12	AN device identifier fourth user code	73A.3	User-defined code value	M	Yes []
AN13	AN message code 10	73A.4	EEE technology message code	M	Yes []